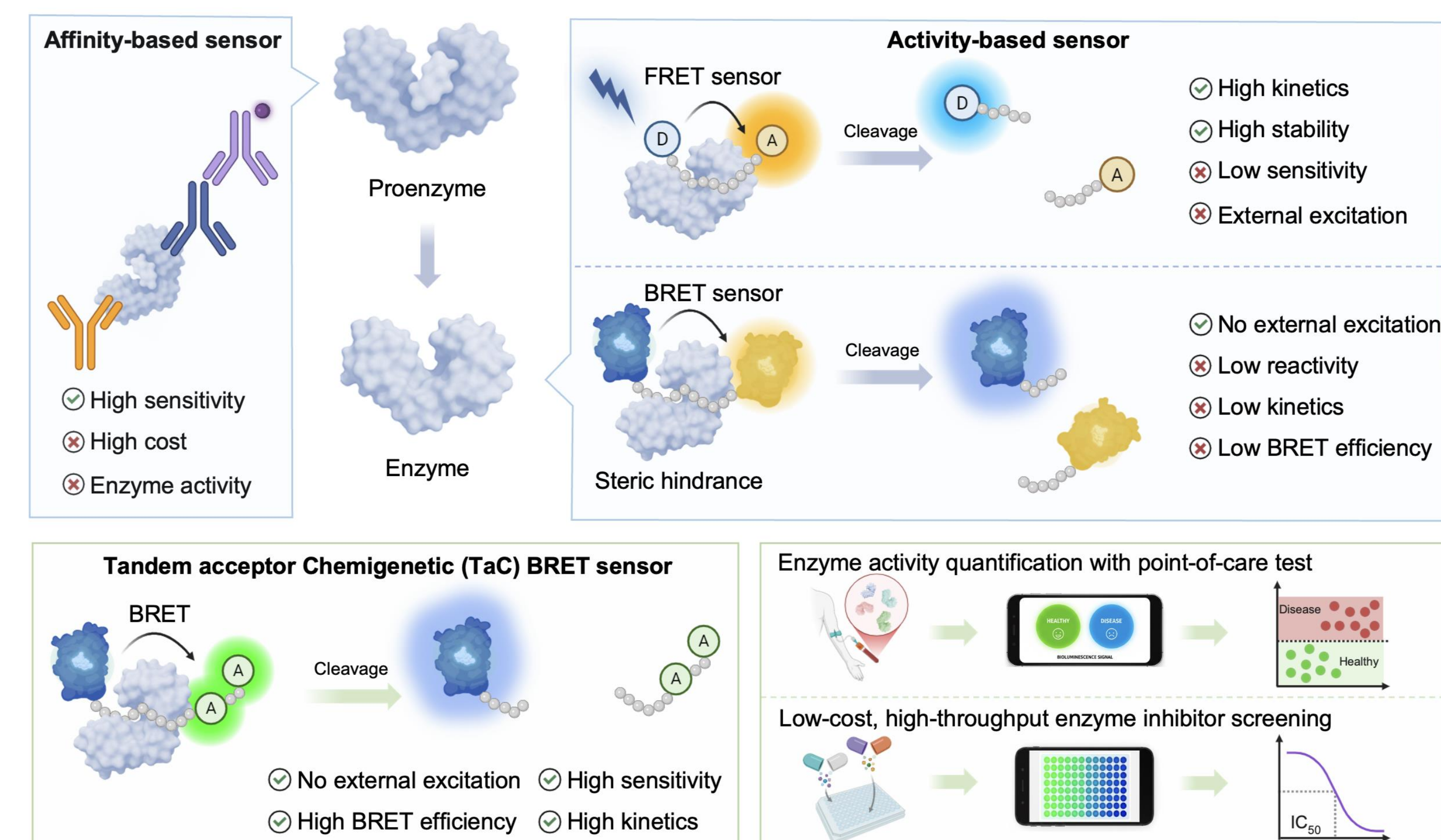
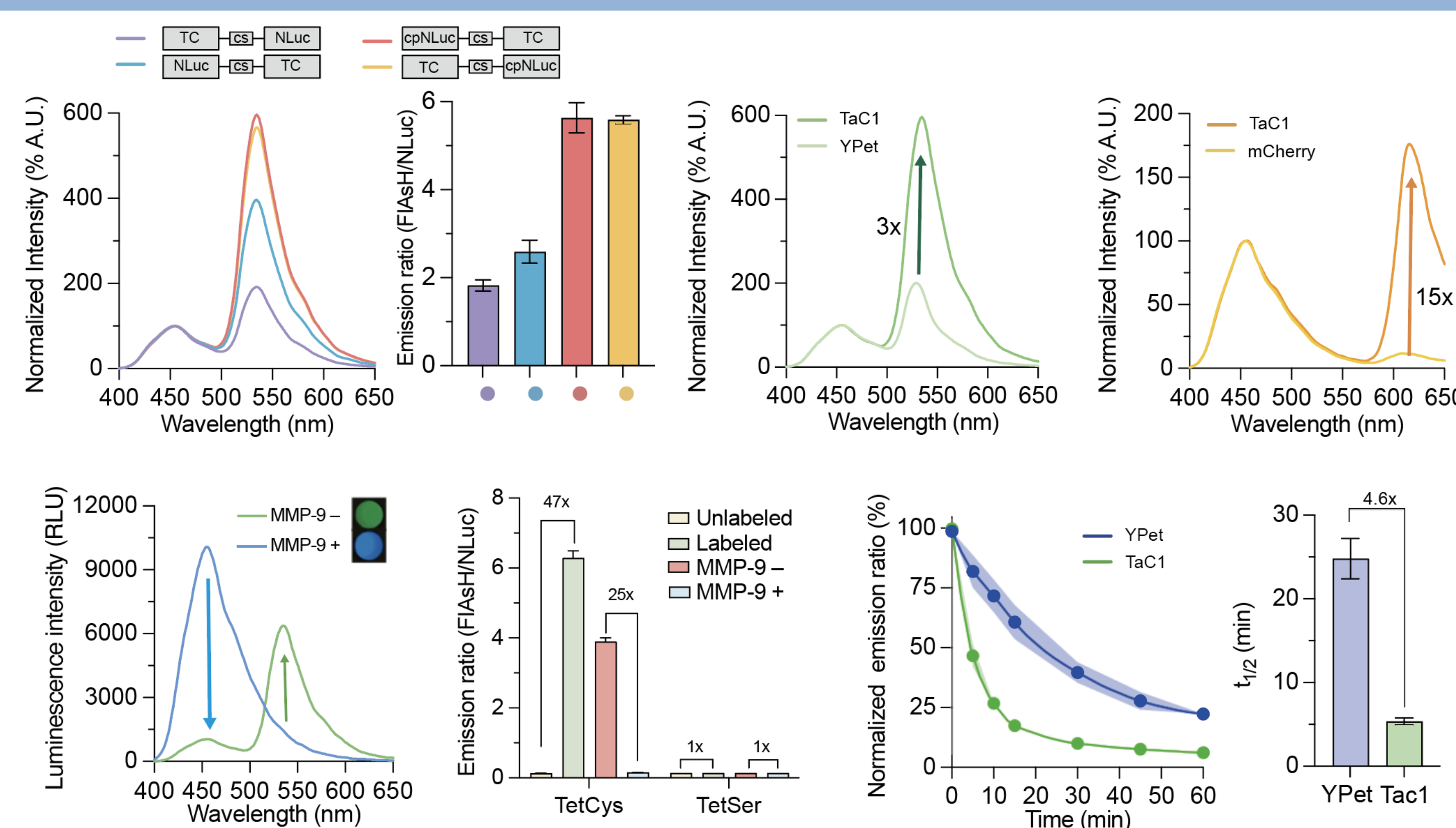
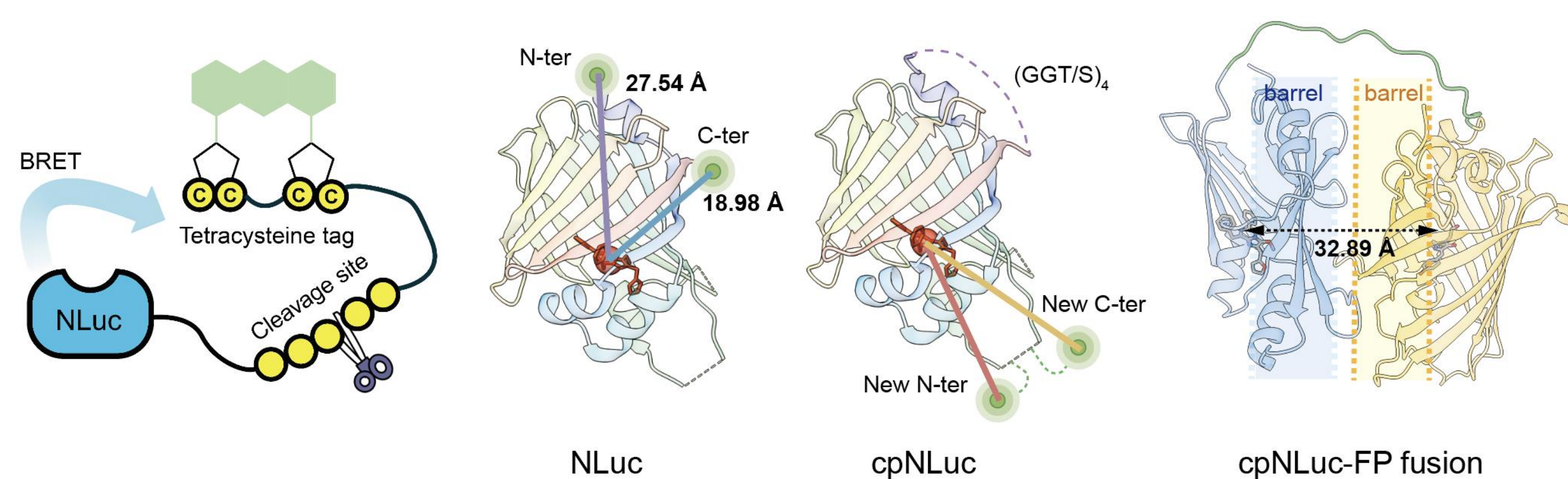


Protease activity is a direct marker of disease, yet existing assays often lack sensitivity and depend on specialized instrumentation, limiting their suitability for point-of-care analysis in clinical samples. Here, we report a modular chemigenetic bioluminescence resonance energy transfer (BRET) platform that integrates a NanoLuc donor with tandem small-molecule dye acceptors for ultrasensitive and quantitative protease detection. The compact acceptor architecture enables near-maximal BRET efficiency (~90%) and unobstructed proteolysis, producing pronounced bioluminescent colour shifts with up to 42-fold ratiometric responses and picomolar detection limits. This plug-and-play design supports diverse protease targets, including MMP-9, MMP-2, thrombin, prostate-specific antigen and NS3/4A, through simple substitution of the substrate peptide. In a pilot clinical study, the MMP-9 sensor enabled minute-scale, smartphone-based quantification of protease activity in human plasma and distinguished patients with colorectal tumours, including those with early adenomatous polyps, with 91.7% diagnostic accuracy. Using the same handheld readout, the MMP-2 sensor enabled low-cost and high-throughput screening of protease inhibitors. Together, these results establish tandem-acceptor chemigenetic (TaC) BRET as a sensitive, portable platform for activity-based protease diagnostics and drug discovery.

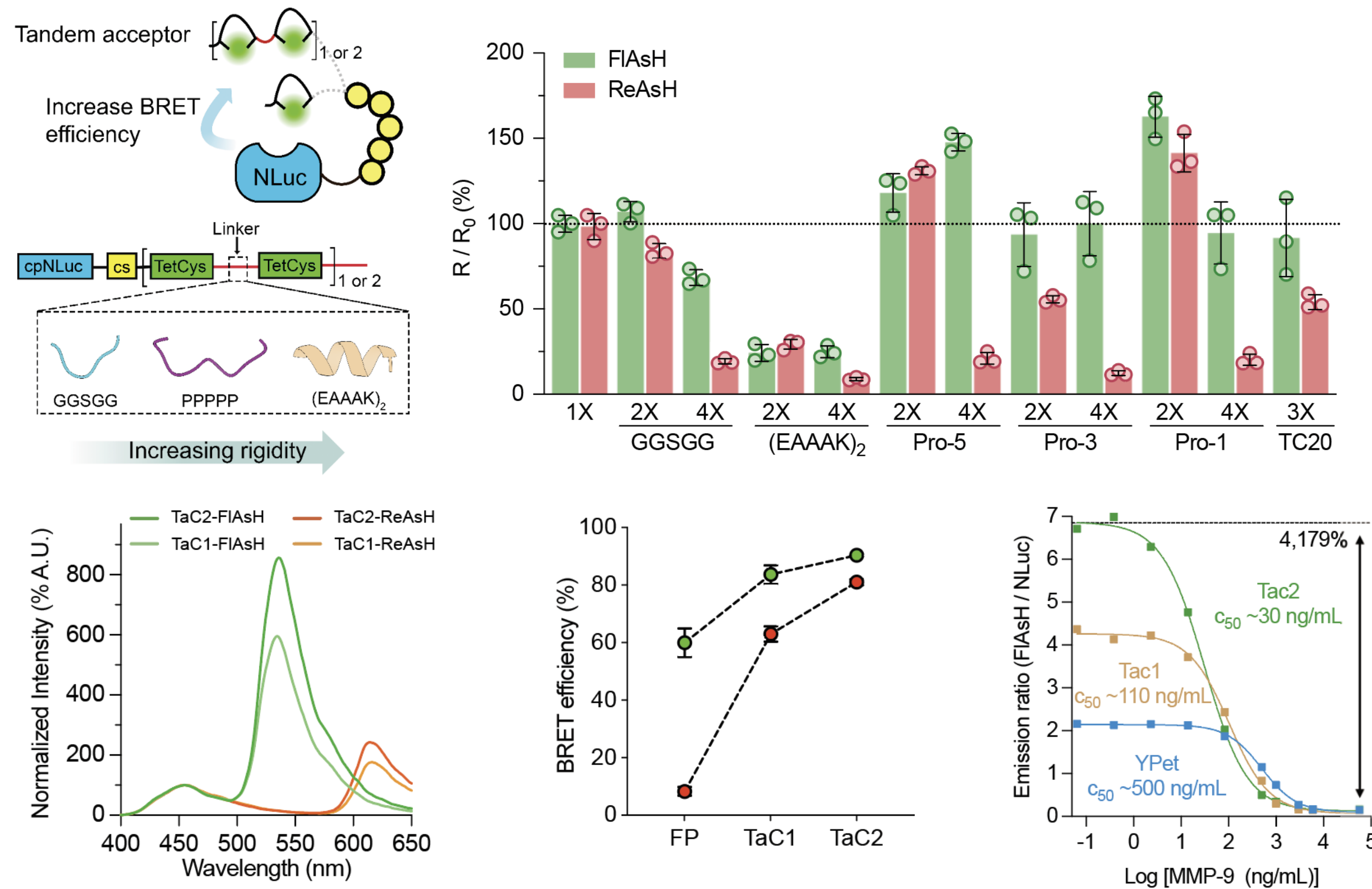


Result

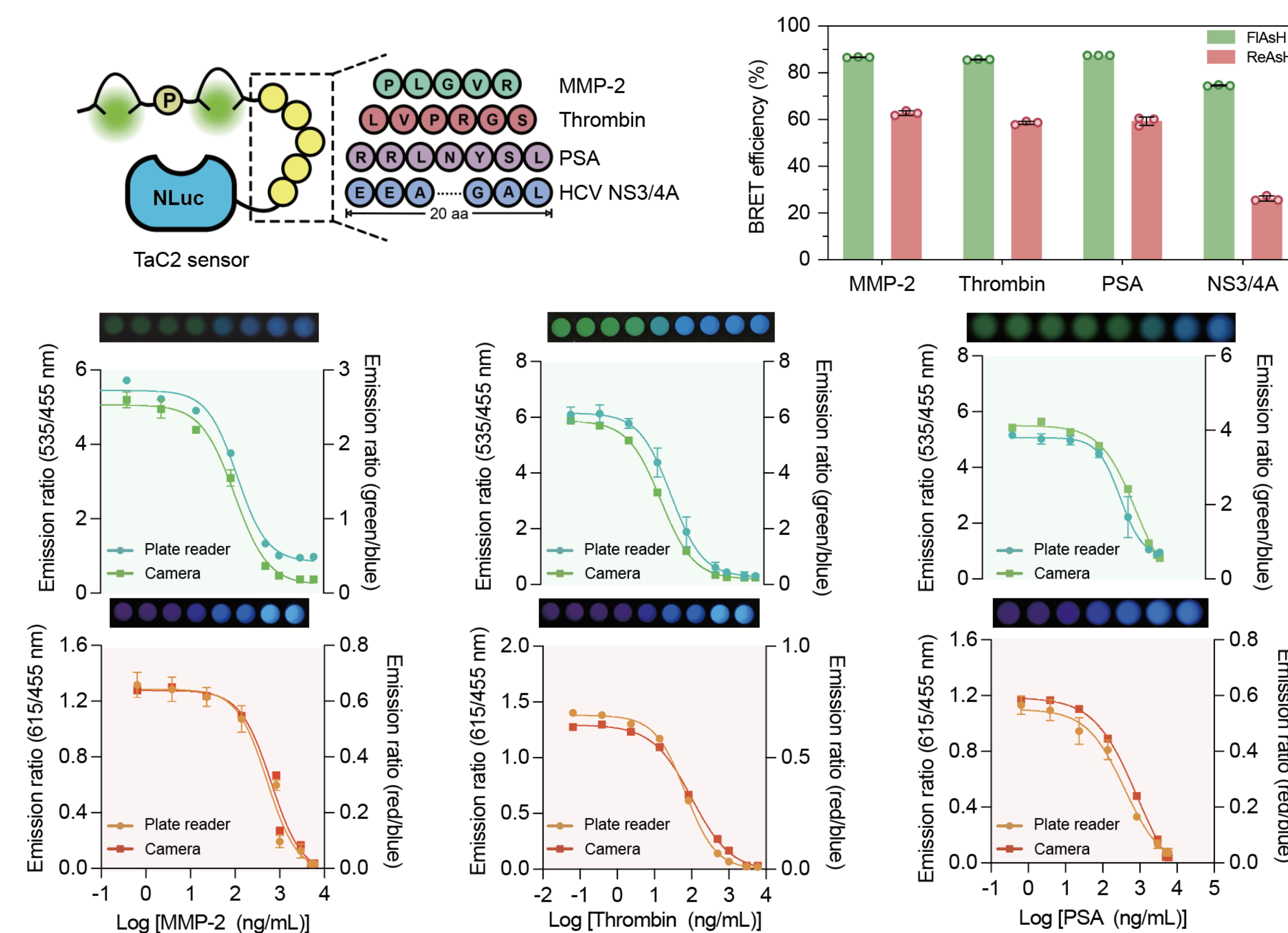
1. Development of a chemigenetic BRET sensor for MMP-9



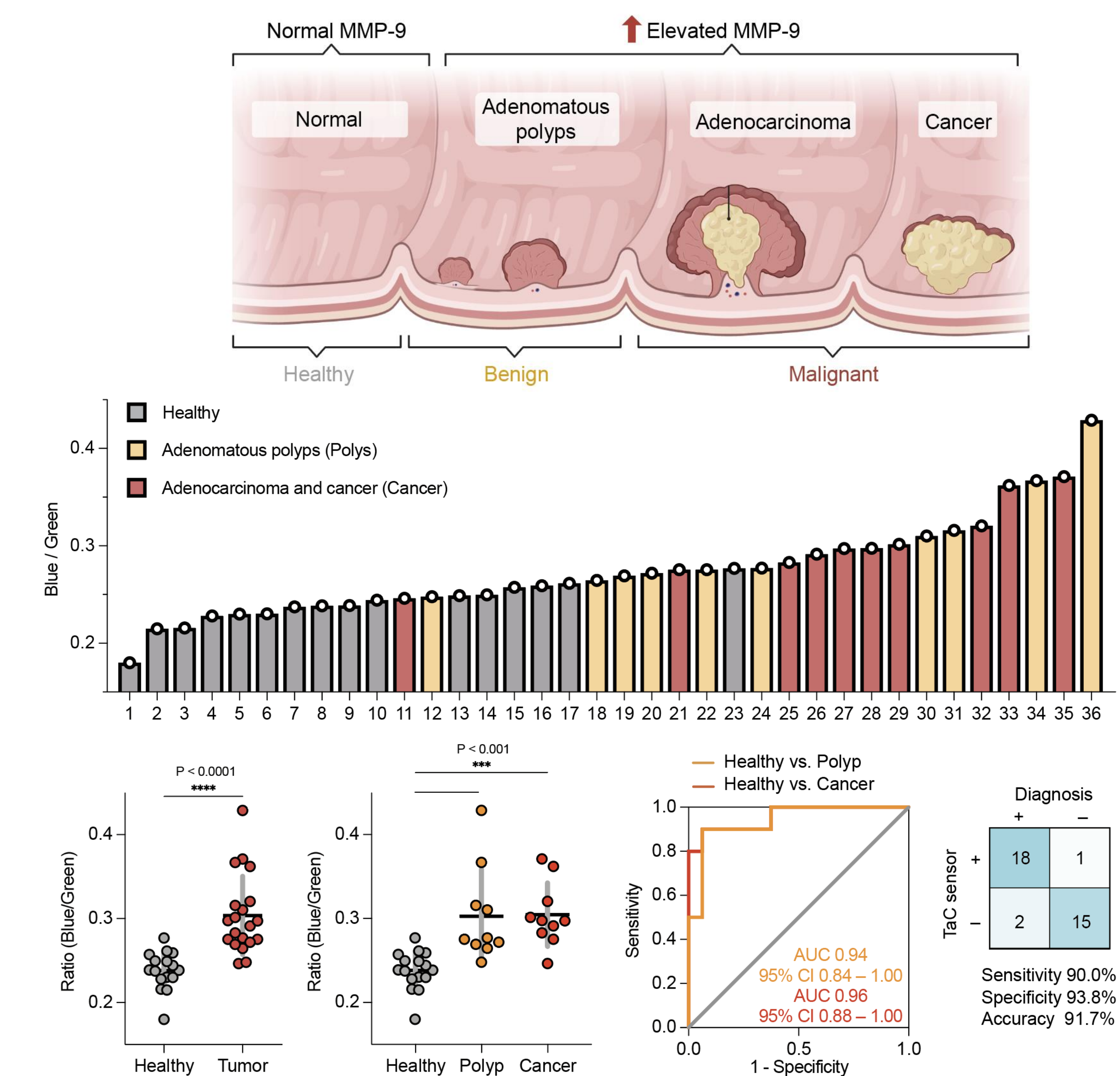
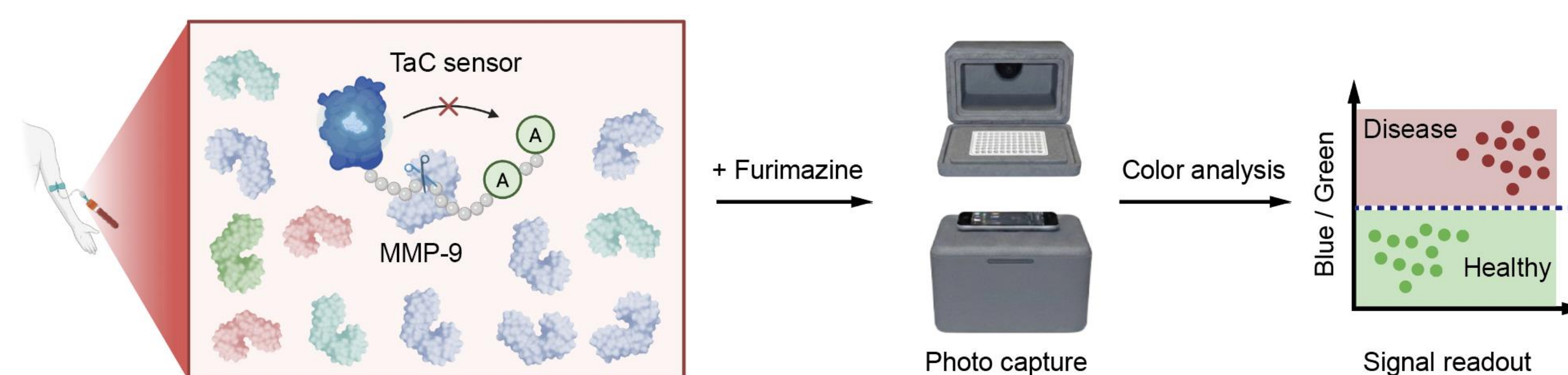
2. Tandem-acceptor BRET sensors with near-quantitative BRET efficiency



3. Smartphone Camera-based TaC for modular protease sensing



4. TaC enables point-of-care diagnosis of colorectal disease



5. TaC enables smartphone-based screening and characterization of MMP-2 inhibitors

